

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

INJONGO's travel domain is derived from CLINC's generic intent set ('exchange rate', 'book flight') [\[Q39\]](#) and explicitly omits intercity bus operator flows. Empirical inspection of the Amharic split confirms that all sampled travel-domain Amharic utterances use air-travel framing (book_flight to Turkey, Sudan) or generic car rental — there are no intercity bus, seat-class, or operator-specific intents [\[DATASET-D7, D8, D25, D26\]](#). The deployment requires intent detection for Sky Bus / Selam Bus / Ride bus and ride-hail booking, including seat-class upgrade, road-closure rebooking, holiday-surge booking, and operator-specific cancellation, none of which appear in the 40-intent taxonomy. The paper itself acknowledges scope limits [\[Q107, Q111\]](#). Although elicitation marked IO as MODERATE priority (user accepts general travel intents as broadly comparable), the empirical ontology mismatch in the travel subdomain is concrete and substantial.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g. " (p.3)
Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ፤ / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D25	intent=book_flight, COUNTRY: ቱርክ (Turkey) — Flight to Turkey — no intercity bus, seat class, or operator-specific slots present
D26	intent=book_flight, COUNTRY: ሱዳን (Sudan) — Flight to Sudan — air travel framing, not surface transport

Strengths

- Joint intent + slot taxonomy is well-defined and matches the deployment's NLU module structure [\[Q2, Q31\]](#)
- Travel is one of five included domains, providing partial overlap with bus booking flows [\[Q5, Q117\]](#)
- Ground-up African-context elicitation rather than translation reduces some Western framing [\[Q8, Q30\]](#)

ID	Evidence
Q2	"Intent detection and slot-filling are crucial components of the natural language understanding module in task-oriented dialogue systems (Hemphill et al., 1990; Coucke et al., 2018; Gupta et al., 2018)." (p.1)
Q5	"We cover the following five domains: banking, home, travel, utility, and kitchen & dining." (p.1)
Q8	"Through extensive experiments, we benchmark the fine-tuning multilingual transformer models and the prompting large language models (LLMs), and show the advantage of leveraging African-cultural utterances over Western-centric utterances for improving cross-lingual transfer from the English language." (p.1)
Q30	"In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)

Q31	"INJONGO is a joint intent detection and slot-filling dataset (ID-SF) for typologically diverse Sub-Saharan African languages and English." (p.2)
Q117	"These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home)." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include intercity bus route queries, fare lookup, seat selection (VIP/economy), booking, cancellation/rebooking — only generic 'book_flight', 'car_rental', and 'travel_suggestion' overlap, established by elicitation and observed in the Amharic data [DATASET-D7, D8].

IO-2: Yes. INJONGO covers 5 domains and 40 intents derived from CLINC [Q39, Q117] and omits intercity bus, seat-class selection, holiday-surge booking, and road-closure rebooking. The paper explicitly notes the taxonomy 'missing some other domains such as healthcare and education' [Q107] and a non-natural intent frequency distribution [Q111].

IO-3: The travel domain is dominated by air travel framing in the Amharic split (international flights to Turkey, Sudan) [DATASET-D7, D8, D25, D26]; these are not necessarily irrelevant but concentrate evaluation weight on air rather than surface intercity transport relevant to the deployment.

IO-4: Documented gaps: (a) no intercity bus operator intents, (b) no seat-class/fare-tier intents, (c) no rebooking-on-disruption intent, (d) no Ethiopian-holiday surge booking intents — all of which harm content validity for this specific deployment.

ID	Evidence
Q39	"The source data for our multilingual benchmark is from the CLINC English dataset—an intent detection with 150 intent classes across 10 domains (but without slot annotation), we extracted 40 intents from five most suitable domains to the African contexts: Banking (e.g. "transfer", "pay bill"), Home (e.g. "play music", "calendar update"), Kitchen and Dining (e.g. "recipe", "confirm reservation"), Travel (e.g. "exchange rate", "book flight"), and Utility (e.g." (p.3)
Q107	"The scope of INJONGO is constrained by its coverage of only 5 domains and 40 intents, missing some other domains such as healthcare and education that are essential for real-world applications." (p.9)
Q111	"Additionally, the fixed distribution of examples across intents may not accurately reflect the natural frequency of these interactions in real-world conversations." (p.9)
Q117	"These are a total of 40 categories across 5 domains (Banking, Kitchen and Dining, Travel, Utility, and Home)." (p.13)
D7	ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንግረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D25	intent=book_flight, COUNTRY: ቱርክ (Turkey) — Flight to Turkey — no intercity bus, seat class, or operator-specific slots present
D26	intent=book_flight, COUNTRY: ሱዳን (Sudan) — Flight to Sudan — air travel framing, not surface transport

Information Gaps

- Whether any Amharic travel-domain utterances in the unsampled 99.4% of the dataset use surface-transport framing

Requires Expert Verification

- Definitive intent inventory of Sky Bus / Selam Bus / Ride booking APIs to quantify gap
-

Remediation

Gap 1: INJONGO's 40-intent travel taxonomy is air-travel framed and lacks intercity bus operator flows (seat-class, rebooking, holiday-surge)

Recommendation: Define a deployment-specific intent supplement (~10–15 intents) drawn from Sky Bus / Selam Bus / Ride API surfaces; collect 50–100 Amharic utterances per intent via native speaker elicitation following INJONGO's methodology; treat this as a required held-out evaluation alongside INJONGO's travel domain.